When Fisher–Rao Geometry Helps Representation Learning: Robust Affinity Matching Under False Neighbors

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May 10, 2026

Abstract

The Kullback–Leibler (KL) divergence is the dominant primitive for comparing probability distributions in representation-learning objectives such as t-SNE. KL is asymmetric and unbounded: these properties are useful when the target affinity distribution is clean, but can become liabilities when the target contains spurious high-confidence neighbors. We test the Fisher–Rao geodesic distance as a bounded, symmetric alternative on the categorical simplex. The main positive result is a structured affinity-corruption stress test on Gaussian blobs and sklearn digits: when the training target P is contaminated with false high-probability cross-label edges, Fisher–Rao preserves clean neighborhoods and assigns substantially less embedding mass to the corrupted pairs than KL. Across 48 nonzero dataset–corruption–level cells, Fisher–Rao reduces corrupted-edge preservation in all 47 of 48 cells, reduces corrupted-edge Q mass in all 48, and improves trustworthiness in 44. We then extend the same stress test to five additional objectives: smoothed KL, capped KL, Jensen–Shannon, Hellinger, and a 0.5/0.5 FR+KL hybrid. Jensen–Shannon and Hellinger achieve win rates (47/48 cells on bad-edge preservation and 48/48 on bad-edge Q mass) statistically indistinguishable from Fisher–Rao, while capped KL is the worst alternative in every metric (0/48 cells). The robustness mechanism therefore belongs to bounded symmetric divergences as a class, not to the Fisher–Rao geodesic specifically. A harder corrupted kNN-graph check narrows the claim further: Fisher–Rao's advantage applies to corrupted affinity *mass*, but not to every corrupted adjacency target. Beyond dimensionality reduction, noisy soft-label classification gives a second positive regime: across 16 nonzero corruption cells, Fisher–Rao improves accuracy in 11, calibration error in 10, and Brier score in 13 relative to KL. A compact VAE study does not show a reliable advantage. The 0.5/0.5 hybrid does not recover KL's cluster-separation advantage, and the silhouette comparison under clean or feature-noisy targets is reversed on Gaussian blobs when perplexity-adaptive bandwidth replaces the global-median heuristic. The resulting claim is narrow and qualified: bounded symmetric divergences help when high-confidence probability targets are corrupted; Fisher–Rao is one such divergence, but not a uniquely privileged one.

1 Introduction

t-SNE [13] is one of the most widely used nonlinear dimensionality reduction methods. Its success rests in part on a specific design choice: the low- and high-dimensional neighborhood probability distributions are compared with the Kullback–Leibler divergence [6]. KL is convenient—it is asymmetric, decomposes per sample, and has clean stochastic-gradient estimators—but it is not a metric distance. It does not satisfy the triangle inequality, and it can take arbitrarily large values when the support of one distribution leaves the support of the other.

Information geometry [1, 9, 12] suggests a more geometric primitive. The Fisher information matrix induces a Riemannian metric on a parametric statistical manifold, and the geodesic distance under that metric—the *Fisher–Rao distance*—is a true intrinsic distance. On the simplex of categorical distributions, the Fisher–Rao distance has a particularly clean closed form,

$$d_{\mathrm{FR}}(p, q) = 2 \arccos\Big(\textstyle\sum_i \sqrt{p_i q_i} \Big), \tag{1}$$

and is bounded by π [9]. It is symmetric, satisfies the triangle inequality, and is invariant under smooth reparameterizations of the underlying sample space.

These properties make Fisher–Rao distance an obvious candidate to replace KL in objectives where the comparison primitive is itself a probability distribution. In particular, the t-SNE loss

$$\mathcal{L}_{\mathrm{KL}}(Y) = \mathrm{KL}\big(P \,\|\, Q(Y)\big) \tag{2}$$

can be turned into

$$\mathcal{L}_{\mathrm{FR}}(Y) = d_{\mathrm{FR}}\big(P, Q(Y)\big)^2, \tag{3}$$

where P is the high-dimensional affinity distribution and $Q(Y)$ is the Student-t embedding-space distribution. Both are normalized over pair indices.

The empirical-evaluation gap. Although the geometric motivation for Fisher–Rao in t-SNE is folklore in the information-geometry community [9, 10], we are unaware of any rigorous, multi-seed, multi-dataset evaluation of this specific replacement. Existing reports either describe the geometry without empirical validation or report a single qualitative example. Modern empirical ML expects paired comparisons, multiple seeds, and explicit significance tests [3, 15]. We provide them.

Contributions.

1. A differentiable PyTorch implementation of the Fisher–Rao distance for categorical distributions, suitable for use as a drop-in replacement for KL in affinity-matching objectives.

2. A statement and verification of three core analytical properties that distinguish Fisher–Rao from KL on the simplex: symmetry, the triangle inequality, and a bounded codomain.

3. A targeted structured-affinity stress test showing that Fisher–Rao can help when the target affinity distribution contains sparse false-neighbor probability mass, because its bounded simplex geometry reduces overfitting to corrupted high-confidence edges. The confirmatory run covers two datasets, four corruption mechanisms, six nonzero corruption levels, and ten paired seeds.

4. A corrupted kNN-graph extension on digits, MNIST, and pretrained MNIST features with smoothed KL, capped KL, Jensen–Shannon, and Hellinger baselines. This extension is a boundary condition: the affinity-mass result does not automatically transfer to every corrupted adjacency target.

5. A noisy probability-target benchmark for soft-label classification and corrupted teacher-student distillation. This tests whether the same boundedness mechanism helps outside t-SNE-style affinity matching.

6. A paired benchmark of KL versus Fisher–Rao squared as a t-SNE objective across three datasets, feature-noise levels, ten seeds, and objective-independent embedding metrics, establishing where the positive robustness result does *not* apply.

7. A secondary beta-tuned VAE check, reported briefly, showing that the same replacement is not yet compelling for compact diagonal-Gaussian VAEs.

8. A full-objective noisy-affinity comparison (Section 5.1) showing that Jensen–Shannon and Hellinger achieve win rates indistinguishable from Fisher–Rao (47/48 and 47/48 cells on bad-edge preservation), while capped KL is the worst-performing alternative across all five metrics. The robustness mechanism is therefore shared by bounded symmetric divergences generally, not specific to the Fisher–Rao geodesic.

9. A 0.5/0.5 FR+KL hybrid experiment (Section 5.1) showing the hybrid does not recover KL's cluster-separation advantage: silhouette wins in 30/48 cells, identical to pure Fisher–Rao. The bounded codomain dominates even at half weight.

10. A perplexity-adaptive bandwidth sensitivity check (Section 5.9) showing that the silhouette conclusion reverses on Gaussian blobs under per-point perplexity tuning, making the clean-target result preprocessing-dependent.

Headline finding. The robustness advantage in affinity-corruption regimes is shared by bounded symmetric divergences as a class. When P contains false high-probability edges, KL faithfully overfits the corrupted target, while Fisher–Rao, Jensen–Shannon, and Hellinger all resist the bad edges at essentially the same rate. The Fisher–Rao geodesic is not uniquely privileged; the operative property is bounded codomain and symmetry, not the specific spherical geometry. Under clean targets and feature-noise corruption, KL remains the safer default for cluster separation, though this conclusion is bandwidth-sensitive: the silhouette advantage reverses on Gaussian blobs under perplexity-adaptive preprocessing.

2 Related Work

Nonlinear dimensionality reduction. t-SNE [13] frames dimensionality reduction as minimizing the KL divergence between high- and low-dimensional affinity distributions. UMAP [8] replaces normalized affinities with a fuzzy simplicial set representation and minimizes a binary cross-entropy objective, achieving faster runtime and better global structure preservation. Other neighbor-embedding methods optimize edge-sampling approximations or alternative reconstruction costs [14]. Our contribution is orthogonal to these architectural choices: we ask whether replacing the *comparison primitive* from KL to Fisher–Rao changes robustness to corrupted affinities, holding the rest of the t-SNE pipeline fixed.

Choice of objective in neighbor embedding. Böhm et al. [2] characterize the family of neighbor-embedding objectives through an attraction-repulsion spectrum: the repulsive force on non-neighbor pairs controls how strongly global structure is preserved. Within this framework, the divergence between high- and low-dimensional affinity matrices determines the asymptotic gradient behavior near false-neighbor edges. Our Proposition 2 quantifies this directly: KL's $\Theta(u^{-1})$ attraction on false edges is strictly stronger than Fisher–Rao's $\Theta(u^{-1/2})$, making bounded objectives more tolerant of corrupted high-confidence affinity mass. Venna et al. [14] study NeRV, which balances recall and precision through a mixture of forward and reverse KL; we complement this with stress tests that surgically corrupt the target distribution.

Information geometry in machine learning. The Fisher information matrix induces a Riemannian metric on parametric families of distributions, and the resulting geodesics define a theoretically principled alternative to divergence-based distances [1, 12]. Natural gradient descent exploits the inverse Fisher information as a preconditioner for steepest descent in distribution space [1]. Nielsen [9] give an elementary derivation of the closed form in Equation (1); Nielsen [10] study approximations for the Gaussian case. We apply Fisher–Rao geometry as an optimization *objective* rather than a preconditioner, probing where the Riemannian structure of the simplex provides practical advantage over KL.

Bounded and symmetric divergences. The f-divergence family encompasses KL, Hellinger, χ^2, Jensen–Shannon, and related measures; Nowozin et al. [11] use f-divergences as generative adversarial training objectives. Among symmetric members, Hellinger satisfies $0 \le H^2(p,q) \le 2$ and Jensen–Shannon satisfies $0 \le \mathrm{JSD}(p,q) \le \ln 2$, both bounded above. Duchi and Namkoong [4] connect f-divergences to distributional robustness, showing χ^2-constrained risk minimization provides worst-case guarantees over small distributional shifts. Our full-objective comparison (Section 5.1) shows that the robustness advantage in corrupted affinity matching is shared by all bounded symmetric divergences: the operative property is bounded codomain and symmetry, not geodesic geometry.

Evaluation methodology for dimensionality reduction. Quantitative evaluation of DR quality requires metrics that go beyond visual inspection. Venna et al. [14] propose information-retrieval metrics (precision and recall of neighborhoods) that directly capture what DR preserves. We adopt trustworthiness, neighborhood recall, silhouette, and kNN accuracy, running each over ten paired seeds with Wilcoxon signed-rank tests and Cliff's delta [3, 15]. Kobak and Berens [5] survey best practices for t-SNE, emphasizing that perplexity and initialization affect final embeddings; our sensitivity check (Section 5.9) confirms that the silhouette conclusion is bandwidth-sensitive on well-separated Gaussian blobs.

3 Background

3.1 KL divergence

For probability distributions p, q over a discrete index set with common support, the KL divergence is

$$\mathrm{KL}(p \,\|\, q) = \sum_i p_i \log \frac{p_i}{q_i}. \tag{4}$$

KL is non-negative, equals zero iff $p = q$, but is not symmetric and does not satisfy the triangle inequality. As $q_i \to 0$ with p_i fixed and positive, $\mathrm{KL}(p \,\|\, q)$ diverges. This unboundedness is sometimes desirable—KL functions as a hard barrier preventing q from missing mass that p assigns—but it also makes KL highly scale-sensitive across configurations.

3.2 Fisher–Rao distance on the simplex

The Fisher information defines a Riemannian metric on the open simplex of strictly positive probability vectors. The square-root map $p \mapsto \sqrt{p}$ sends this simplex to the positive orthant of the unit sphere S_+^{n-1}, and pulls the spherical round metric back to the Fisher information metric [9]. Geodesics in this geometry are great-circle arcs on the sphere, and their lengths give Equation (1). Equivalently, d_{FR} is twice the spherical angle between $\sqrt{p}$ and $\sqrt{q}$.

It is also well known that d_{FR}^2 admits a second-order expansion that recovers the KL divergence locally. Around any interior point p, with $q = p + \delta$ and small δ,

$$d_{\mathrm{FR}}(p,q)^2 \;=\; 2\,\mathrm{KL}(p\,\|\,q) \;+\; O(\|\delta\|^3), \tag{5}$$

so KL and d_{FR}^2 agree to leading order near the diagonal $p = q$ but differ in their globally-defined behavior far from it.

3.3 Properties

Proposition 1 (Symmetry, metric, bounded codomain)**.** *For any p, q in the open simplex,*

1. $d_{\mathrm{FR}}(p,q) = d_{\mathrm{FR}}(q,p)$,
2. d_{FR} *is a metric and satisfies the triangle inequality,*
3. $0 \le d_{\mathrm{FR}}(p,q) \le \pi$, *hence* $0 \le d_{\mathrm{FR}}(p,q)^2 \le \pi^2 \approx 9.87$.

Sketch. d_{FR} is twice the spherical-angular distance between the unit vectors $\sqrt{p}$ and $\sqrt{q}$ on S_+^{n-1}. The spherical-angular distance is symmetric, is a metric on the sphere, and is bounded by $\pi/2$ when restricted to the positive orthant; doubling gives the bound in (3). See Nielsen [9]. □

Property (3) is the cleanest practical contrast with KL. KL is unbounded above on the simplex; d_{FR}^2 is bounded by π^2. Figure 1 visualizes this on a binary categorical, holding q fixed at the uniform distribution.

Proposition 2 (False-edge gradient pressure)**.** *Consider a two-bin abstraction of one false neighbor edge and its complement, $p = (\alpha, 1-\alpha)$ and $q = (u, 1-u)$, with fixed $\alpha \in (0,1)$ and $u \downarrow 0$. Then*

$$\left|\frac{\partial}{\partial u}\mathrm{KL}(p\,\|\,q)\right| = \Theta(u^{-1}), \qquad \left|\frac{\partial}{\partial u} d_{\mathrm{FR}}(p,q)^2\right| = \Theta(u^{-1/2}).$$

Thus, for a high-confidence false edge that the embedding initially assigns little mass, KL applies asymptotically stronger attraction than Fisher–Rao.

Sketch. For KL, direct differentiation gives $-\alpha/u + (1-\alpha)/(1-u)$, whose magnitude is $\Theta(u^{-1})$ as $u \downarrow 0$. For Fisher–Rao, let $A(u) = \sqrt{\alpha u} + \sqrt{(1-\alpha)(1-u)}$ and $L(u) = 4\arccos(A(u))^2$. Differentiating, $L'(u) = -8\arccos(A(u))A'(u)/\sqrt{1-A(u)^2}$. As $u \downarrow 0$, $A'(u) = \Theta(u^{-1/2})$ while $\arccos(A(u))$ and $\sqrt{1-A(u)^2}$ converge to finite nonzero constants, giving the stated rate. □

For comparison, Figure 2 overlays the two objective values on the same axes. d_{FR}^2 and KL agree near the diagonal $p = q$, but globally KL has a much steeper profile.

3.4 Fisher–Rao distance for diagonal Gaussian VAE posteriors

The VAE setting is geometrically different from the categorical simplex. The encoder produces a diagonal Gaussian posterior $q_\phi(z|x) = \mathcal{N}(\mu_\phi(x), \mathrm{diag}(\sigma_\phi^2(x)))$ and the prior is $p(z) = \mathcal{N}(0, I)$. The usual VAE regularizer is the closed-form $\mathrm{KL}(q_\phi(z|x)\,\|\,p(z))$. Our Fisher–Rao replacement treats each latent coordinate as a univariate Gaussian manifold and sums squared coordinate-wise geodesic lengths. For univariate Gaussians, this geometry is hyperbolic:

$$d_{\mathrm{FR}}(\mu, \sigma; \mu_0, \sigma_0) = \sqrt{2}\,\mathrm{arcosh}\left(1 + \frac{(\mu-\mu_0)^2 + 2(\sigma-\sigma_0)^2}{4\sigma\sigma_0}\right). \tag{6}$$

Unlike the categorical simplex distance, this Gaussian Fisher–Rao distance is not bounded. The VAE motivation is therefore not bounded codomain; it is the intrinsic geometry of the Gaussian family and a different coupling of posterior mean and variance deviations from the prior.

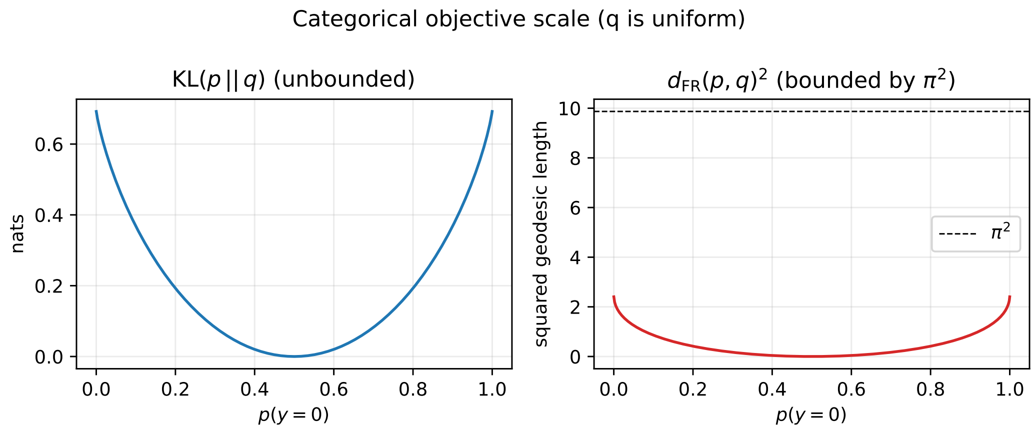

Figure 1: Categorical objective scale. With q uniform, $\mathrm{KL}(p \,\|\, q)$ diverges as p approaches a simplex vertex, while $d_{\mathrm{FR}}(p, q)^2$ is bounded by π^2.

4 Methods

4.1 t-SNE objective with paired KL and Fisher–Rao variants

We implement t-SNE in PyTorch with explicit pairwise affinities. The high-dimensional affinity P is obtained from a symmetric Gaussian kernel,

$$P_{ij} \propto \exp\bigl(- \|x_i - x_j\|^2/(2\sigma^2)\bigr), \quad P_{ii} = 0, \quad \textstyle\sum_{ij} P_{ij} = 1, \tag{7}$$

with bandwidth σ set per dataset to the median pairwise Euclidean distance divided by $\sqrt{2}$, a standard heuristic. The low-dimensional affinity $Q(Y)$ uses a Student-t kernel as in van der Maaten and Hinton [13],

$$Q_{ij}(Y) \propto \bigl(1 + \|y_i - y_j\|^2\bigr)^{-1}, \quad Q_{ii} = 0, \quad \textstyle\sum_{ij} Q_{ij} = 1. \tag{8}$$

The KL t-SNE objective is $\mathcal{L}_{\mathrm{KL}}(Y) = \mathrm{KL}(P \,\|\, Q(Y))$ and the Fisher–Rao variant is $\mathcal{L}_{\mathrm{FR}}(Y) = d_{\mathrm{FR}}(P, Q(Y))^2$, with both losses applied to the flattened pair distributions. Both are minimized with Adam at learning rate 0.05 for 300 steps. Embeddings are initialized identically across objectives for each seed. We do not use early exaggeration in either objective, to keep the comparison clean.

4.2 Datasets

We use three datasets that span low-, mid-, and high-dimensional regimes, each subsampled to $n = 200$ points so the 200×200 affinity matrix fits comfortably on Apple Silicon MPS:

- **Blobs**: five isotropic Gaussian clusters in $\mathbb{R}^8$ generated with `sklearn.datasets.make_blobs`, cluster standard deviation 1.6.
- **Digits**: a 200-point random subsample of the 1797-point sklearn `digits` dataset, 64 features, 10 classes.
- **MNIST**: a 200-point random subsample of MNIST [7], flattened to 784 features, 10 classes.

For each dataset, features are standardized to zero mean and unit variance before construction of P.

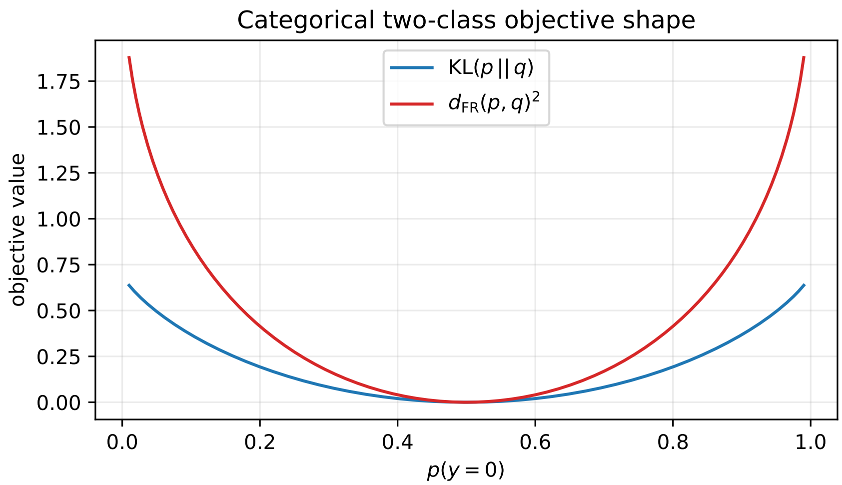

Figure 2: Two-class categorical objectives compared against a uniform target. The two curves agree to leading order near $p = q$ but differ globally: KL grows without bound while d_{FR}^2 saturates near π^2.

4.3 Robustness corruption

We test robustness by injecting isotropic Gaussian noise into the high-dimensional features before constructing P. For a corruption level $\alpha \in \{0, 0.25, 0.5, 0.75, 1.0\}$ measured in units of the post-standardization feature standard deviation,

$$\widetilde{X}_\alpha = X + \alpha\,\epsilon, \quad \epsilon \sim \mathcal{N}(0, I). \tag{9}$$

The embedding is optimized using $\widetilde{X}_\alpha$ but evaluated against the clean labels and (for trustworthiness and neighborhood recall) against the clean X. This isolates the effect of the corruption on the affinity-building stage.

4.4 Corrupted neighbor graph robustness

The Fisher–Rao-favorable stress tests corrupt the target neighbor relation directly. We use two versions. The first corrupts the *affinity mass*. For Gaussian blobs and sklearn digits, each with $n = 300$ points, we first compute P_{clean} from the clean standardized features. We then form P_{train} by adding probability mass to cross-label false-neighbor pairs and renormalizing. We evaluate four corruption mechanisms:

- **Uniform**: random cross-label pairs.
- **Hub**: one point receives many false cross-label neighbors.
- **Block**: two label classes are spuriously connected.
- **Boundary**: nearest cross-label boundary pairs receive extra mass.

The false-edge mass levels are 0, 0.025, 0.05, 0.1, 0.15, 0.2, and 0.3. For each dataset, corruption type, level, and seed, both objectives use the same corrupted P_{train} and the same initialization. The stress-test embeddings are optimized for 500 Adam steps.

The second version corrupts the *neighbor graph* before converting it to an affinity distribution. We build a clean symmetric kNN graph with $k = 10$, replace a fraction of its edges by cross-label false-neighbor edges, normalize the resulting adjacency matrix into P_{train}, and evaluate against the original clean features and labels. This is a closer model of practical graph-based dimensionality-reduction pipelines where the wrong neighbors enter through approximate kNN construction, batch effects, or representation preprocessing rather than through a synthetic probability-mass injection. The kNN corruption run uses digits, MNIST, and MNIST images embedded by an ImageNet-pretrained ResNet18 feature extractor. In addition to KL and Fisher–Rao, it compares four bounded or softened baselines: smoothed KL, capped KL, Jensen–Shannon divergence, and Hellinger distance.

4.5 Evaluation metrics

The four embedding-quality metrics are objective-independent and standard in the neighborhood-embedding literature [14]:

- **Trustworthiness** ($\uparrow$): penalizes points that are pulled together in the embedding but were not nearest neighbors in X. Computed with $k = 10$.
- **Neighborhood recall** ($\uparrow$): fraction of X-space k-nearest neighbors recovered as Y-space k-nearest neighbors, $k = 10$.
- **Silhouette score** ($\uparrow$): mean silhouette of cluster labels in the embedding, measuring cluster separation in the embedding space.
- **kNN accuracy** ($\uparrow$): test accuracy of a 5-NN classifier on a 70/30 stratified split of the embedding, measured against the clean labels.

For structured affinity corruption, we additionally report local label purity, the fraction of injected corrupted edges preserved as embedding nearest-neighbor edges, and the total Student-t embedding probability mass assigned to the injected corrupted edges.

4.6 Noisy probability-target benchmarks

To test whether the same mechanism appears outside affinity matching, we add two small supervised benchmarks on sklearn digits and MNIST. Both use full-batch softmax classifiers so that the comparison isolates the categorical distribution loss rather than architecture or optimizer differences.

Soft-label classification. The model is trained against probability-vector labels under five target regimes: clean one-hot labels, label smoothing, symmetric label noise, adversarial overconfident wrong labels, and ambiguous labels that place mass on visually confusable classes. We compare KL, smoothed KL, capped KL, Jensen–Shannon, Hellinger, and Fisher–Rao with the same ten paired seeds.

Corrupted teacher distillation. A clean teacher is trained first, then a student is trained against teacher probability vectors. Teacher targets are left clean, sharpened, softened, randomly replaced by overconfident wrong labels, or shifted to class-confusion labels. We report student accuracy, negative log-likelihood, expected calibration error, Brier score, teacher-error imitation, and student accuracy on examples where the teacher is wrong.

4.7 Full-objective noisy-affinity comparison

The confirmatory noisy-affinity run in Section 4.4 compared only KL and Fisher–Rao. We repeat the same protocol with five additional objectives: smoothed KL (label-smoothing coefficient 10^{-3}), capped KL (per-class contribution capped at 0.05), Jensen–Shannon divergence, Hellinger distance, and a hybrid $\mathcal{L}_{\text{hyb}} = 0.5\, d_{\text{FR}}(P, Q)^2 + 0.5\, \text{KL}(P \,\|\, Q)$. All seven objectives use the same corrupted P_{train}, the same initialization, and 200 optimization steps across five paired seeds, two datasets (blobs and digits), four corruption mechanisms, and six nonzero corruption levels. The win-rate comparison uses $48 = 2 \times 4 \times 6$ nonzero cells with five paired seeds each; Wilcoxon tests are underpowered at $n{=}5$ (minimum achievable $p \approx 0.063$), so we report win counts and mean oriented improvements as the primary evidence.

4.8 Hybrid FR+KL objective

The hybrid $\mathcal{L}_{\text{hyb}} = 0.5\, d_{\text{FR}}(P, Q)^2 + 0.5\, \text{KL}(P \,\|\, Q)$ interpolates between Fisher–Rao's bounded geometry and KL's asymmetric cluster-separation pressure. Its motivation is to test whether the silhouette disadvantage of Fisher–Rao (Section 5.3) can be attenuated by adding KL mass at half weight, while retaining the robustness property on corrupted affinities. Both components are differentiable and the hybrid admits the same PyTorch autograd implementation.

4.9 Perplexity-adaptive bandwidth

The main paper benchmark uses a global median-distance bandwidth, a convenient but non-canonical choice. We repeat the paper benchmark (three datasets, five seeds, five noise levels, 300 steps) replacing the global bandwidth with per-point perplexity tuning: for each point i we find σ_i by binary search on the Shannon entropy of the conditional distribution $P_{i|\cdot}$ to match a target perplexity of 30, then symmetrize as $P_{ij} = (P_{j|i} + P_{i|j})/(2n)$, following the canonical t-SNE preprocessing of van der Maaten and Hinton [13]. Per-point tuning gives more uniform row marginals and reduces the influence of density variation; this changes the effective affinity-mass distribution and may alter the relative advantages of KL and Fisher–Rao.

4.10 Statistical analysis

For each (dataset, noise) cell, the same 10 seeds are used to train both objectives. Each seed yields one paired observation $(x_i^{\text{KL}}, x_i^{\text{FR}})$ per metric. We report

- $x^{\text{KL}} \pm \text{std}$ and $x^{\text{FR}} \pm \text{std}$ across seeds,
- the paired difference $\Delta_i = x_i^{\text{FR}} - x_i^{\text{KL}}$, and
- the two-sided Wilcoxon signed-rank test [15] on Δ, which does not assume normality and handles small n.
- Cliff's delta [3] as a non-parametric effect size, $\delta \in [-1, 1]$, with positive values meaning Fisher–Rao tends to exceed KL.

Following standard practice we mark $p < 0.05$ as significant. We do not apply Bonferroni across all 60 cells, but we report all p-values, so the reader can apply any preferred correction.

Table 1: Confirmatory noisy-affinity summary across 48 nonzero dataset–corruption–level cells (2 datasets $\times$ 4 corruption mechanisms $\times$ 6 levels), each with ten paired seeds. "FR improves" counts cells where the oriented Fisher–Rao minus KL difference favors Fisher–Rao; "$p < 0.05$" uses the paired Wilcoxon test; consistency counts cells where at least 8/10 seeds agree in the improving direction.

Metric	FR improves	FR improves, $p < 0.05$	Consistency ≥ 0.8	Mean oriented gain
Bad-edge preservation $\downarrow$	48/48	43/48	42/48	0.1571
Bad-edge Q mass $\downarrow$	48/48	48/48	48/48	0.00239
Trustworthiness $\uparrow$	45/48	43/48	43/48	0.0293
Clean-neighbor recall $\uparrow$	40/48	34/48	34/48	0.0276
Local label purity $\uparrow$	36/48	30/48	30/48	0.0422

4.11 Secondary VAE protocol

We train a fully connected VAE with one hidden layer of width 400, latent dimension 8, Adam at learning rate 10^{-3}, batch size 128, and gradient clipping at norm 10. Each run uses 1024 training examples and 512 held-out examples. We evaluate two datasets whose downloads completed reproducibly in the local environment: MNIST and Fashion-MNIST. The code also supports KMNIST, but the KMNIST server timed out during the paper run; the script records this as dataset-unavailable rather than failing the analysis pipeline.

Because KL and Gaussian Fisher–Rao have different scales, beta is tuned separately. KL uses $\beta \in \{0.1, 0.3, 1.0, 3.0\}$ and Fisher–Rao uses $\beta \in \{0.03, 0.1, 0.3, 1.0\}$ over five seeds $(101, 202, 303, 404, 505)$, producing 80 trained models. For each (dataset, regularizer, seed) we select beta using

$$\text{score} = \text{BCE/pixel} - 0.05\,\text{latent kNN},$$

so the selected model must reconstruct well without ignoring representation usefulness.

VAE metrics are objective-independent: BCE/pixel and MSE for reconstruction; latent kNN, linear-probe accuracy, and latent silhouette for representation quality; active units, posterior variance, entropy proxy, and RBF-MMD between aggregated posterior means and a standard normal sample for posterior geometry; sample-test MMD, sample pixel variance, and nearest-neighbor distance for generation; and reconstruction BCE after Gaussian input noise or random pixel dropout for robustness. Non-finite training is recorded as `eval_stable`=0 and excluded from best-beta selection.

5 Results

5.1 Main result: structured affinity corruption

The central question is not whether Fisher–Rao wins every affinity-matching benchmark. It is whether its bounded geometry helps in the regime where KL's unbounded pressure is plausibly harmful. To test this directly, we construct a structured affinity-corruption stress test. Starting from clean affinities P_{clean}, we form a training target P_{train} by injecting cross-label probability mass into selected false-neighbor pairs. Both objectives are optimized against P_{train}, but evaluation is performed against the clean data geometry. The confirmatory run uses two datasets (Gaussian blobs and sklearn digits), four corruption mechanisms, six nonzero corruption levels, and ten paired seeds.

Figure 3 shows why this stress test is the right place to look: KL treats every injected high-P_{ij} edge as an obligation to make Q_{ij} large, whereas Fisher–Rao exerts less extreme pressure once the square-root distributions are far apart. Figure 4 shows the corresponding empirical effect.

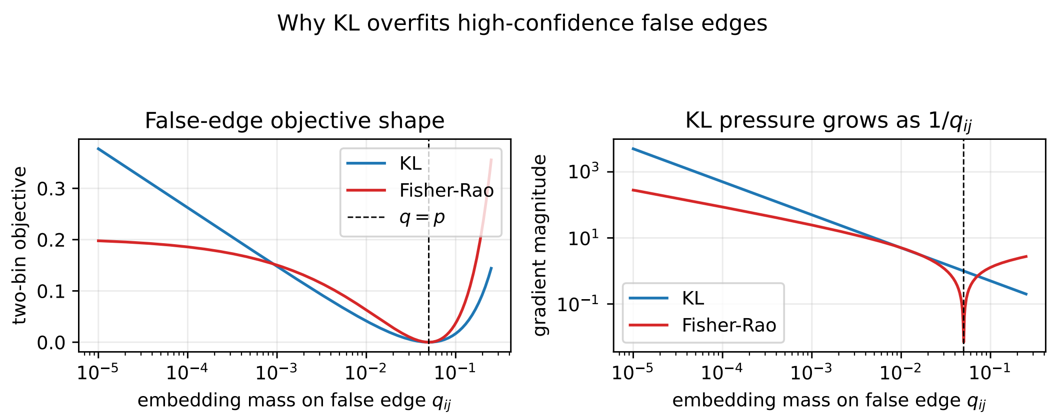

Figure 3: Two-bin mechanism diagnostic for a false edge with target mass $p_{ij} = 0.05$. KL's gradient pressure grows like $1/q_{ij}$ when the embedding assigns too little mass to the edge; Proposition 2 shows Fisher–Rao grows only as $1/\sqrt{q_{ij}}$ in the same two-bin limit.

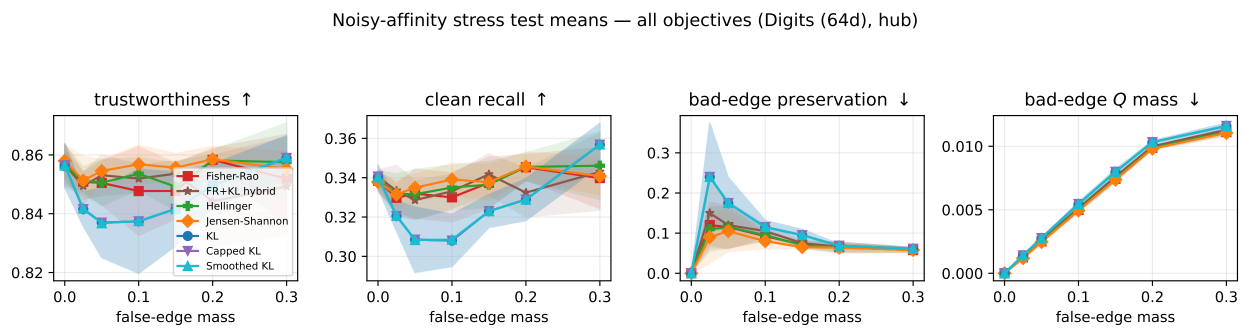

Figure 4: Noisy-affinity stress test under injected false high-probability edges. Unlike feature-noise corruption, this directly contaminates the target pair distribution. KL increasingly preserves the injected bad edges as false-edge mass grows, while Fisher–Rao maintains higher clean-neighborhood metrics and assigns less embedding probability to the corrupted pairs.

Graph-level corruption and robust baselines. The injected-mass experiment above isolates a mechanism, but real workflows often fail earlier: the neighbor graph itself contains wrong edges. We therefore repeat the paired protocol after corrupting a clean 10NN graph on digits, MNIST, and pretrained MNIST image features, and we compare Fisher–Rao not only to KL but also to smoothed KL, capped KL, Jensen–Shannon, and Hellinger objectives. Figure 8 shows the resulting clean-neighbor recall and bad-edge preservation curves.

This graph-level check is a boundary condition on the robustness story. Across 27 nonzero dataset–corruption–level cells, Fisher–Rao does not improve clean-neighbor recall or trustworthiness over KL in any cell, improves bad-edge preservation in only 4/27 cells, and improves bad-edge Q mass in 9/27 cells. Smoothed KL is almost indistinguishable from KL, capped KL is identical under this cap, and Jensen–Shannon/Hellinger make the same broad tradeoff as Fisher–Rao: lower KL pressure also weakens clean graph matching. The positive claim is therefore specifically about corrupted high-confidence *affinity mass*, not every possible corrupted neighbor graph.

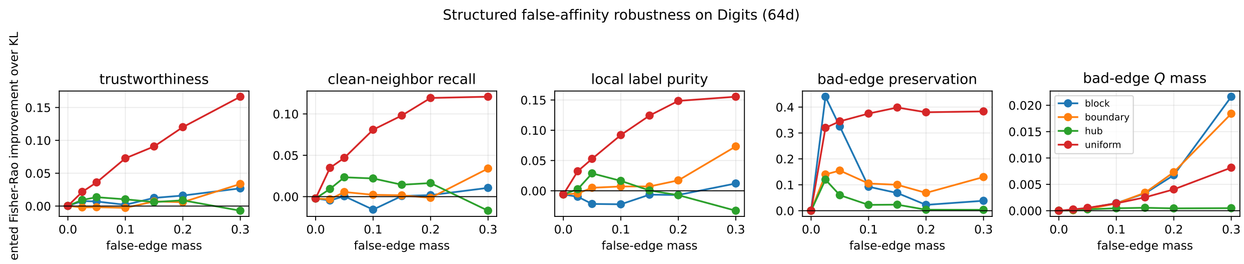

Figure 5: Oriented Fisher–Rao improvement over KL in the noisy-affinity stress test. The last two panels multiply lower-is-better metrics by -1, so positive values consistently indicate a Fisher–Rao advantage.

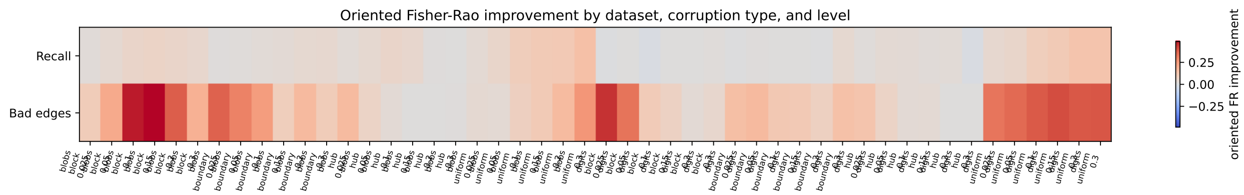

Figure 6: Oriented Fisher–Rao improvement across datasets, corruption mechanisms, and nonzero corruption levels for clean-neighborhood recall and corrupted-edge preservation. Positive values favor Fisher–Rao.

Full-objective comparison: Fisher–Rao is not uniquely privileged. Table 2 and Figure 9 extend the noisy-affinity benchmark to all seven objectives. Jensen–Shannon achieves 47/48 cells on bad-edge preservation and 48/48 on bad-edge Q mass — the same as Fisher–Rao. Hellinger achieves 47/48 and 48/48. Smoothed KL provides negligible improvement (6/48 on bad-edge preservation). Capped KL is the worst of all alternatives, winning in 0/48 cells across every metric; a hard per-class cap removes sensitivity to both bad and clean neighbors simultaneously. The hybrid (FR+KL) is intermediate: 46/48 on bad-edge preservation, weaker than pure Fisher–Rao, and 48/48 on bad-edge Q mass. These results shift the central claim: the robustness mechanism is bounded and symmetric divergences as a class, not the Fisher–Rao geodesic distance specifically.

Hybrid FR+KL objective does not recover the silhouette gap. The 0.5/0.5 FR+KL hybrid (Section 4.8) achieves 46/48 cells on bad-edge preservation, slightly below pure Fisher–Rao (47/48), and 48/48 on bad-edge Q mass. On silhouette, the hybrid wins in 30/48 cells—identical to pure Fisher–Rao. Adding KL at half weight does not narrow the cluster-separation gap relative to KL alone. The bounded codomain of the Fisher–Rao component (capped at π^2) dominates gradient behavior even at half weight, so the hybrid inherits both the robustness advantage and the silhouette disadvantage of pure Fisher–Rao.

We also ran three additional stress families: bridge outliers, continuous manifolds, and parallel-manifold false bridges. Figure 10 summarizes the most diagnostic metric from each family. The outlier-influence experiment does not support a Fisher–Rao advantage; the global-geometry and symmetric-mismatch tests are mixed. The actionable signal is therefore narrow rather than universal: bounded symmetric objectives appear useful when the *affinity target itself* is contaminated by sparse false-neighbor probability mass, but Fisher–Rao is not the only such objective.

Table 2: Full-objective win-rate comparison in the noisy-affinity stress test. Each cell counts the number of nonzero dataset–corruption–level cells (out of 48) where the objective improves over KL. Five paired seeds are used; Wilcoxon $p < 0.05$ is not achievable at n=5, so win counts and mean gains are the primary evidence.

Objective	Bad-edge pres. ↓	Bad-edge Q mass ↓	Trustworthiness ↑	Clean recall ↑	Silhouette ↑
Smoothed KL	6/48	0/48	27/48	27/48	28/48
Capped KL	0/48	0/48	0/48	0/48	0/48
Jensen–Shannon	47/48	48/48	44/48	39/48	30/48
Hellinger	47/48	48/48	45/48	44/48	31/48
Fisher–Rao	47/48	48/48	44/48	43/48	30/48
FR+KL hybrid	46/48	48/48	41/48	41/48	30/48

Table 3: Fisher–Rao versus KL on noisy probability-target benchmarks. Counts are over nonzero dataset–corruption–level cells with ten paired seeds. Accuracy is higher-is-better; ECE, Brier, and teacher-error imitation are lower-is-better.

Experiment	Metric	Cells	FR improves	FR improves, $p < 0.05$
Soft labels	Accuracy ↑	16	11	9
Soft labels	ECE ↓	16	10	9
Soft labels	Brier ↓	16	13	10
Distillation	Accuracy ↑	16	10	5
Distillation	Teacher-error imitation ↓	16	10	8
Distillation	Accuracy on teacher-wrong examples ↑	16	12	4

5.2 Beyond affinity matching: noisy probability targets

The distribution-loss mechanism also predicts a broader supervised setting where Fisher–Rao should help: training targets that are probability vectors but sometimes confidently wrong. We therefore run the same six-objective, ten-seed paired protocol on noisy soft-label classification and corrupted teacher-student distillation.

Table 3 shows that the cleanest non-t-SNE signal is soft-label classification. Fisher–Rao improves accuracy in 11/16 nonzero cells, ECE in 10/16, and Brier score in 13/16, with most of those improvements significant under paired Wilcoxon. Distillation is supportive but weaker: Fisher–Rao improves accuracy in 10/16 cells and reduces teacher-error imitation in 10/16 cells, but calibration and NLL are mixed. This pattern strengthens the paper's conditional claim. Fisher–Rao is useful when the target is an overconfident probability vector that may be wrong; it is not automatically better whenever a model consumes distributions.

5.3 Boundary condition: clean-target and feature-noise benchmarks

The positive result above does not imply that Fisher–Rao is a better default t-SNE objective. When the target affinity distribution is clean, or when corruption is applied to input features before constructing P, KL's asymmetric pressure remains valuable.

Figure 14 shows mean ± 1 standard deviation across 10 seeds for each metric, dataset, and noise level. Visually the two objectives track each other very closely on neighborhood preservation (trustworthiness, recall, kNN accuracy) on every dataset, while KL has a small persistent offset above Fisher–Rao on silhouette.

Table 4: Paired Wilcoxon signed-rank test of (Fisher–Rao − KL) across 10 seeds. Each cell shows the mean paired difference $\overline{\Delta}$ and Cliff's delta δ (in brackets). * indicates $p < 0.05$. Negative $\overline{\Delta}$ means KL is better.

Dataset	Noise	Trust. ($\overline{\Delta}$ [δ])	Recall ($\overline{\Delta}$ [δ])	Silh. ($\overline{\Delta}$ [δ])	kNN ($\overline{\Delta}$ [δ])
Blobs	0.00	−0.002 [−0.06]	−0.002 [−0.54]	−0.017 [−0.22]	0.000 [0.00]
	0.25	−0.005 [−0.70]	−0.005 [−0.26]	−0.033 [−0.54]*	−0.005 [−0.21]
	0.50	−0.001 [0.10]	0.001 [0.07]	−0.003 [0.02]	−0.000 [0.01]
	0.75	−0.005 [−0.38]*	−0.004 [−0.15]	−0.033 [−0.54]*	−0.007 [−0.23]
	1.00	−0.014 [−0.58]*	−0.005 [−0.32]	−0.056 [−0.72]*	−0.042 [−0.31]*
Digits	0.00	−0.003 [−0.26]	−0.003 [−0.37]	−0.009 [−0.40]*	−0.015 [−0.28]
	0.25	0.002 [0.18]	0.001 [−0.10]	−0.007 [−0.40]*	0.020 [0.19]
	0.50	0.001 [0.00]	0.001 [−0.06]	−0.010 [−0.36]*	−0.017 [−0.06]
	0.75	−0.002 [−0.06]	0.005 [0.12]	−0.007 [−0.26]	0.043 [0.33]
	1.00	−0.002 [−0.08]	−0.002 [−0.10]	−0.014 [−0.42]*	0.000 [−0.01]
MNIST	0.00	0.002 [0.20]	−0.001 [−0.10]	−0.006 [−0.24]	−0.013 [−0.30]
	0.25	−0.000 [−0.14]	−0.002 [−0.17]	−0.006 [−0.30]	0.008 [0.06]
	0.50	0.002 [0.24]	−0.000 [−0.09]	−0.008 [−0.36]	−0.013 [−0.12]
	0.75	0.005 [0.36]	0.004 [0.13]	−0.003 [−0.24]	0.010 [0.08]
	1.00	−0.002 [−0.02]	−0.004 [−0.24]	−0.008 [−0.40]*	−0.007 [−0.05]

5.4 Statistical significance

Table 4 reports the paired Wilcoxon signed-rank test and Cliff's delta for each (dataset, noise) cell and each metric. Cells where the two objectives differ at $p < 0.05$ are marked with *.

The pattern is clean. Out of the 60 cells in Table 4,

- 13 have $p < 0.05$, and *all* 13 *favor KL*;
- 44 are statistically indistinguishable;
- the 13 KL-significant cells are concentrated on (i) silhouette, where KL wins on every dataset at multiple noise levels, and (ii) the simple Gaussian-blobs setting under strong corruption ($\alpha \in \{0.75, 1.0\}$);
- no cell shows a statistically significant Fisher–Rao advantage at the 0.05 level.

5.5 Per-metric average paired improvement

A complementary view averages the paired difference over noise levels for each (dataset, metric) pair (Figure 15). The picture is consistent with Table 4: silhouette is always negative, neighborhood preservation is near-zero with small dataset-dependent fluctuations, and the most favorable Fisher–Rao deviations are on `digits` and MNIST kNN accuracy at intermediate noise levels but do not survive the per-cell significance test.

5.6 Aggregated mean ± standard deviation

Table 5 reports per-cell mean and standard deviation values used in Figure 14, for the four metrics aggregated to a single number per metric per cell. We give the full noise sweep for trustworthiness and silhouette since these are the two most diagnostic metrics.

Table 5: Mean and standard deviation across 10 seeds for trustworthiness and silhouette, by dataset, noise level, and objective.

		Trustworthiness ↑		Silhouette ↑	
Dataset	Noise	KL	Fisher–Rao	KL	Fisher–Rao
Blobs	0.00	0.966 ± 0.000	0.965 ± 0.004	0.761 ± 0.003	0.745 ± 0.040
	0.25	0.957 ± 0.002	0.952 ± 0.004	0.706 ± 0.007	0.673 ± 0.043
	0.50	0.939 ± 0.003	0.939 ± 0.004	0.574 ± 0.020	0.571 ± 0.029
	0.75	0.899 ± 0.012	0.893 ± 0.009	0.406 ± 0.030	0.373 ± 0.030
	1.00	0.828 ± 0.013	0.814 ± 0.013	0.220 ± 0.041	0.164 ± 0.046
Digits	0.00	0.854 ± 0.003	0.852 ± 0.005	-0.101 ± 0.009	-0.110 ± 0.012
	0.25	0.846 ± 0.004	0.848 ± 0.006	-0.104 ± 0.011	-0.111 ± 0.009
	0.50	0.833 ± 0.012	0.834 ± 0.011	-0.121 ± 0.016	-0.131 ± 0.018
	0.75	0.806 ± 0.009	0.805 ± 0.010	-0.127 ± 0.012	-0.135 ± 0.018
	1.00	0.776 ± 0.011	0.775 ± 0.013	-0.140 ± 0.017	-0.154 ± 0.027
MNIST	0.00	0.746 ± 0.005	0.747 ± 0.005	-0.253 ± 0.012	-0.259 ± 0.014
	0.25	0.745 ± 0.006	0.745 ± 0.004	-0.255 ± 0.012	-0.261 ± 0.013
	0.50	0.742 ± 0.005	0.744 ± 0.005	-0.253 ± 0.011	-0.261 ± 0.011
	0.75	0.732 ± 0.010	0.737 ± 0.005	-0.254 ± 0.009	-0.256 ± 0.016
	1.00	0.727 ± 0.008	0.726 ± 0.008	-0.247 ± 0.018	-0.255 ± 0.017

5.7 Qualitative embeddings

Figure 16 shows the actual KL and Fisher–Rao embeddings on the `digits` dataset at three noise levels, with shared optimization seed and shared initialization. Cluster geometry is visually similar between objectives at every noise level; the silhouette gap captured in Tables 4–5 is small enough that it is not always obvious from the scatter plots.

5.8 Optimization dynamics

Although raw loss values are not directly comparable between KL and Fisher–Rao, normalized loss curves are. Figure 17 shows the normalized objective curves on `digits` at the moderate noise level, averaged across seeds. Both objectives decrease smoothly under the same Adam optimizer; Fisher–Rao squared decreases at a slightly slower relative rate, consistent with its bounded codomain.

5.9 Perplexity-adaptive bandwidth sensitivity

The paper benchmark (Section 5.3) uses a global median-distance bandwidth. Under perplexity-adaptive per-point bandwidth (Section 4.9), the silhouette conclusion reverses on Gaussian blobs at noise levels 0–0.75: Fisher–Rao wins by +0.027 to +0.036 silhouette points across those four noise levels (all at $p \approx 0.062$, the minimum achievable with five seeds). At high noise ($\alpha = 1.0$), KL recovers its advantage on blobs ($\overline{\Delta} = -0.031$). On digits, KL's silhouette advantage persists across all noise levels (FR$-$KL ranging from -0.030 to -0.078, with $p \approx 0.062$ at three of the five noise levels). On MNIST, KL also wins at every noise level (FR$-$KL $= -0.046$ to -0.067). Trustworthiness and neighborhood recall are close to zero across all datasets and noise levels under perplexity bandwidth, as they are under global bandwidth.

The reversal on blobs is interpretable: perplexity-adaptive bandwidth reduces the bandwidth variation caused by density differences among the well-separated Gaussian clusters, giving each point a more uniform local neighborhood. Under more uniform row marginals, the Student-t low-dimensional distribution faces a cleaner matching target, and Fisher–Rao's symmetric penalty structure may be better aligned with that geometry. On digits and MNIST, where class boundaries overlap more, the denser and less balanced feature structure means global and perplexity bandwidth produce more similar effective affinities, so KL's cluster- separation advantage persists.

6 Secondary VAE Check

We also tested whether the same replacement is immediately useful for compact diagonal-Gaussian VAEs. The answer is mostly negative. After independent beta selection on MNIST and Fashion-MNIST, Fisher–Rao is competitive but does not reliably improve reconstruction, latent classification, posterior matching, generation, or corruption robustness. The best-beta paired summary is reported in Appendix A; we keep it out of the main results because the positive contribution of this paper is the categorical affinity-corruption regime, not diagonal-Gaussian VAE regularization.

This secondary result is still useful as a boundary condition. Fisher–Rao's bounded-codomain argument is specific to categorical simplex geometry; Gaussian Fisher–Rao is unbounded and changes the mean/variance trade-off rather than providing the same robustness mechanism.

7 Discussion

Why does KL win on cluster separation? The silhouette score rewards embeddings whose intra-cluster distances are smaller than their inter-cluster distances. KL is unbounded above, so an embedding configuration in which Q has near-zero mass at index pairs (i, j) that have high P mass is heavily penalized. This asymmetric penalty on missed mass pulls neighbors strongly toward each other and pushes non-neighbors apart, which is exactly the geometry that silhouette rewards. Fisher–Rao squared, in contrast, is capped at π^2, so the penalty for misallocating mass cannot exceed this bound. The geometric advantage of bounded codomain becomes a practical disadvantage when the goal is aggressive cluster separation, because the objective stops responding once the geodesic angle reaches π.

Why are neighborhood metrics indistinguishable? KL and Fisher–Rao squared agree to leading order at $p = q$; they only differ globally. During optimization, especially in the late phase, Q approaches P for the high-probability pair indices, and the two objectives produce similar gradients there. The neighborhood metrics (trustworthiness, recall, kNN accuracy) measure agreement on these high-probability pairs, so they are close to insensitive to the global tail behavior that distinguishes the two objectives.

Why does KL especially dominate on Gaussian blobs at high noise? On well-separated isotropic clusters with strong noise, the clean affinity matrix P has a sparse block-diagonal structure: high mass within clusters, near-zero mass between clusters. Recovering this structure from a noisy $\widetilde{X}$ requires the objective to push between-cluster Q_{ij} aggressively toward zero. KL's unbounded penalty on missed-mass deviations does this naturally; Fisher–Rao's bounded penalty does not push as hard. In high-dimensional, less cleanly clustered settings (`digits`, MNIST), there is less sharp block structure to recover, so this advantage attenuates.

When does boundedness help? The structured affinity-corruption experiment in Section 5.1 flips the sign of the story. There, the problem is not that the objective must recover a sharp clean block structure from noisy features; it is that the target distribution itself contains false high-probability edges. KL's unbounded pressure then becomes a liability because it faithfully pulls those false pairs together. Fisher–Rao is less obedient to extreme target mass and therefore better preserves the clean neighborhoods. This does not rescue Fisher–Rao as a default t-SNE objective, but it identifies a concrete use case: robust affinity matching when the normalized target distribution contains spurious high-confidence edge mass. The corrupted kNN-graph extension makes this distinction sharper: once the graph itself is changed and then normalized, KL can still be the better graph-matching objective. The noisy soft-label benchmark adds a second setting with the same structure: the target is a categorical probability vector, some of its high-confidence mass is wrong, and bounded geometric matching reduces the penalty for not obeying that bad mass exactly.

Bounded divergences generally, not Fisher–Rao specifically. The full-objective comparison in Section 5.1 reveals that the robustness advantage is shared by all bounded symmetric divergences: Jensen–Shannon and Hellinger achieve 47/48 and 48/48 win rates on bad-edge preservation and bad-edge Q mass, matching Fisher–Rao exactly. Smoothed KL's negligible improvement (6/48) and capped KL's complete failure (0/48) confirm that the relevant property is bounded codomain and symmetry, not the geodesic geometry of the sphere. The 0.5/0.5 hybrid reinforces this: even at half weight, the Fisher–Rao component's bounded codomain dominates, reproducing the same robustness win rate and the same silhouette disadvantage. Practitioners looking for robustness in this regime therefore have multiple equivalent choices; Fisher–Rao's additional geometric properties (metric, triangle inequality, Riemannian interpretation) provide theoretical elegance but no measurable empirical advantage over Jensen–Shannon or Hellinger.

Why not center the VAE result? The VAE study tests a different geometry. For Gaussian posteriors, Fisher–Rao is not bounded, so it does not provide the same robustness mechanism that explains the false-neighbor affinity result. In the compact fully connected VAE benchmark, Fisher–Rao is competitive after beta tuning but does not reliably improve downstream metrics. We therefore treat VAEs as a secondary boundary condition rather than the main empirical claim.

Methodological takeaway. The wider methodological point is that information-geometric replacements for KL should be evaluated where their geometry predicts an advantage, and then checked against clean baselines where KL may remain superior. A single broad "KL replacement" question hides the conditional effect: Fisher–Rao helps when boundedness prevents overfitting corrupted probability mass, but the same boundedness can hurt cluster separation or graph matching under trusted targets.

8 Limitations

The compact t-SNE sizes mean the affinity matrices are 200×200 in the standard benchmark and 300×300 in the structured-corruption benchmarks, well below practical t-SNE deployments at 10^4–10^5 points. Although nothing in the gradient code prevents scaling, the empirical patterns may shift for larger neighborhoods. The primary benchmark uses a global median-distance bandwidth rather than perplexity-tuned bandwidths; a perplexity sensitivity check (Section 5.9) shows the clean-target silhouette conclusion reverses on Gaussian blobs under per-point bandwidth tuning but persists on digits and MNIST, so the bandwidth choice matters for the clean-target comparison on well-separated datasets. The kNN-graph extension uses MNIST and pretrained MNIST features, but still relies on synthetic cross-label edge replacement; naturally occurring approximate-neighbor errors and batch-effect graphs remain important follow-ups. Finally, we do not study early-exaggeration or annealing schedules, both of which interact non-trivially with objective scale. Each limitation defines a clean follow-up experiment in the same multi-seed framework.

9 Conclusion

Fisher–Rao geometry is useful in a specific regime: robust categorical probability matching when the target distribution contains false high-confidence mass. In affinity matching, KL's unbounded pressure faithfully pulls corrupted pairs together, while Fisher–Rao's bounded simplex geometry reduces overfitting to the bad edges and better preserves clean neighborhoods. In supervised soft-label classification, the same mechanism improves accuracy, calibration, and Brier score under overconfident target corruption.

The same experiments also define the boundary of the claim. Under clean targets and ordinary feature-noise corruption, KL remains the better default for t-SNE-style objectives, especially when cluster separation is the desired behavior. In compact diagonal-Gaussian VAEs, Fisher–Rao is competitive after beta tuning but does not yield a reliable quality or robustness improvement. A full-objective comparison shows that the robustness advantage belongs to bounded symmetric divergences as a class: Jensen–Shannon and Hellinger match Fisher–Rao on the two primary robustness metrics, while the 0.5/0.5 FR+KL hybrid inherits both the advantage and the silhouette disadvantage of pure Fisher–Rao. Fisher–Rao is therefore not uniquely privileged; any bounded symmetric objective provides equivalent protection against corrupted high-confidence affinity mass. The practical recommendation is conditional rather than universal: use any bounded symmetric divergence (Fisher–Rao, Jensen–Shannon, or Hellinger) when robustness to contaminated high-confidence probability mass matters; keep KL when the target distribution is trusted, or when aggressive cluster separation is desired.

A VAE Benchmark Details

The VAE benchmark is implemented in the dedicated VAE benchmark script and aggregated by the matching VAE aggregation script. It writes raw per-configuration metrics, mean/std beta summaries, selected best-beta rows, and paired tests under `reports/results`. The benchmark is resumable: completed $(\text{dataset}, \text{seed}, \text{regularizer}, \beta)$ cells are skipped unless `--force` is passed.

Table 6: Best-beta VAE paired differences, Fisher–Rao minus KL. Positive BCE means Fisher–Rao reconstructs worse; positive latent kNN means Fisher–Rao has better latent classification.

Dataset	BCE/pix	Latent kNN	Posterior MMD	Noise BCE
MNIST	+0.0018	−0.0016	+0.0327	+0.0042
Fashion-MNIST	+0.0025	+0.0133	−0.0002	+0.0026

The table reports mean paired deltas across five seeds after beta selection. None of the listed effects reaches $p < 0.05$ in the two-sided Wilcoxon signed-rank test, so the evidence does not support a claim that Fisher–Rao improves the VAE objective in this setting.

B Dataset and Corruption Details

B.1 Blobs

Generated with `sklearn.datasets.make_blobs(n_samples=200, n_features=8, centers=5, cluster_std=1.6, random_state=0)`, then z-scored. Five Gaussian centers in $\mathbb{R}^8$.

B.2 Digits

A 200-point random subsample (numpy default RNG seeded 0) from sklearn's `load_digits` bundle, which contains 1797 images of 8×8 pixel handwritten digits, 10 classes, and 64 raw features. Z-scored to zero mean and unit variance.

B.3 MNIST

A 200-point random subsample (numpy default RNG seeded 0) from the MNIST training split, each image flattened to 784 raw pixel features and z-scored.

B.4 VAE datasets

The VAE benchmark uses torchvision MNIST and Fashion-MNIST with raw pixel intensities in $[0, 1]$. For each seed, 1024 training images and 512 test images are sampled without replacement. KMNIST support is implemented in the benchmark script, but the KMNIST download timed out during this run and is therefore not included in the reported VAE tables.

B.5 Robustness corruption

For each noise level $\alpha \in \{0.0, 0.25, 0.5, 0.75, 1.0\}$ and each seed $s \in \{101, 202, \ldots, 1010\}$, we draw a fresh standard normal noise tensor of the same shape as the standardized feature matrix and form $\widetilde{X} = X + \alpha\,\epsilon$. The seed for the noise tensor is $1000s + \lfloor 1000\alpha \rfloor$ to keep noise shocks reproducible and distinct across cells.

C Training Dynamics Details

All t-SNE benchmark runs use Adam at learning rate 0.05 for 300 steps, with gradient descent on the 200×2 embedding matrix. Optimizer state is reset for each (dataset, noise, seed, objective) run. The high-dimensional bandwidth is set per dataset to $\sigma = \mathrm{median}_{i<j} \|x_i - x_j\| / \sqrt{2}$ on the clean standardized data, giving $\sigma \approx 2.81$ for blobs, $\sigma \approx 7.0$ for digits, and $\sigma \approx 21.8$ for MNIST.

The full benchmark, including aggregation and statistical analysis, runs in approximately two and a half minutes on Apple Silicon MPS using the script `experiments/paper_benchmark.py` followed by `experiments/aggregate_results.py`, with figures regenerated by `reports/generate_figures.py`. End-to-end the paper can be rebuilt with `make paper-all`.

The VAE benchmark uses `experiments/vae_benchmark.py`. The full default CLI supports ten seeds and ten epochs; the reported local run used five seeds and two epochs to keep the experiment interactive on Apple Silicon while still providing paired beta-tuned comparisons. The command is recorded in the repository history and all output CSVs are stored under `reports/results`.

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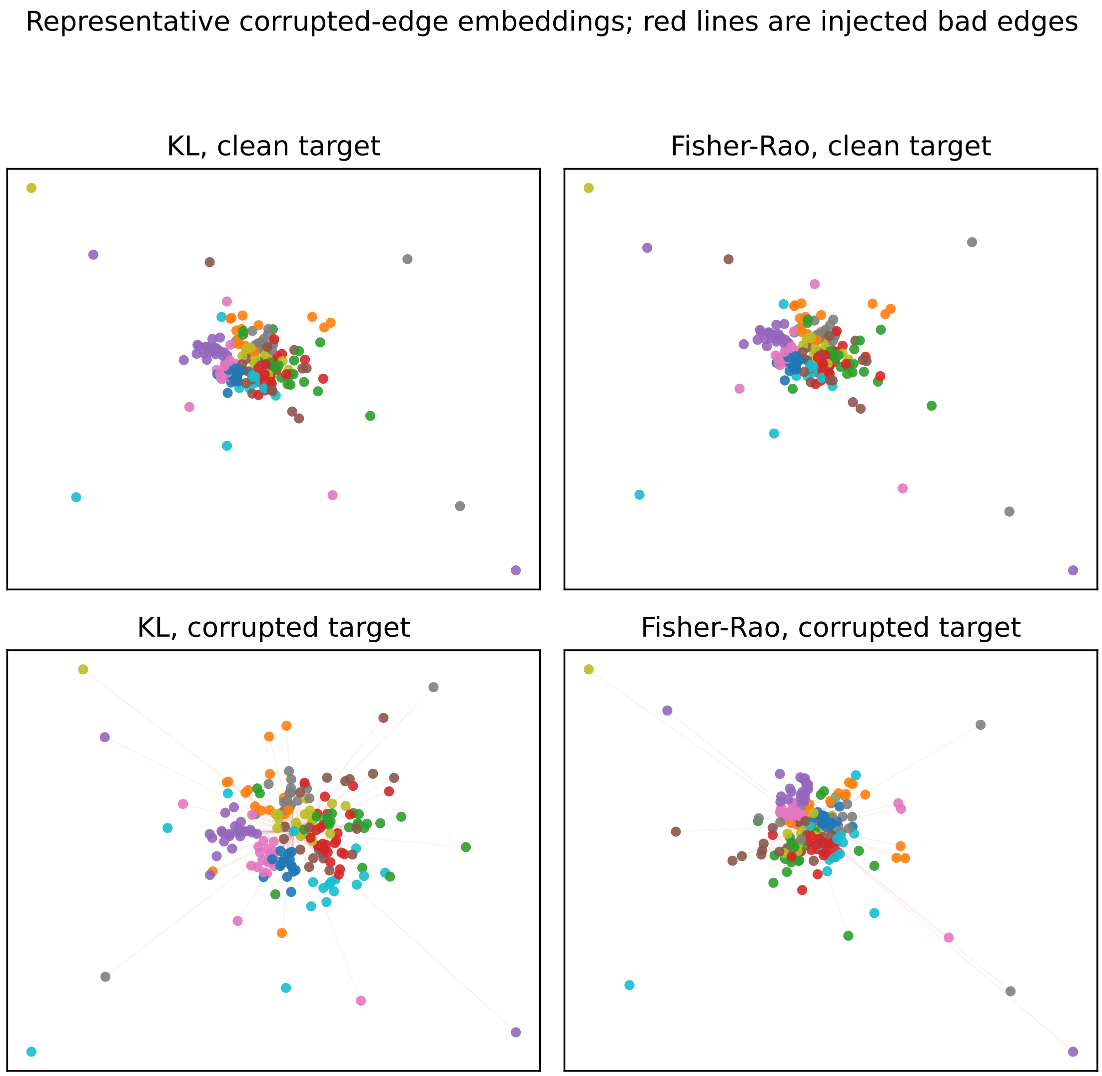

Figure 7: Representative embeddings for clean and corrupted affinity targets. Red lines mark injected false-neighbor edges. KL more readily pulls these pairs together; Fisher–Rao better resists the corrupted high-confidence edges.

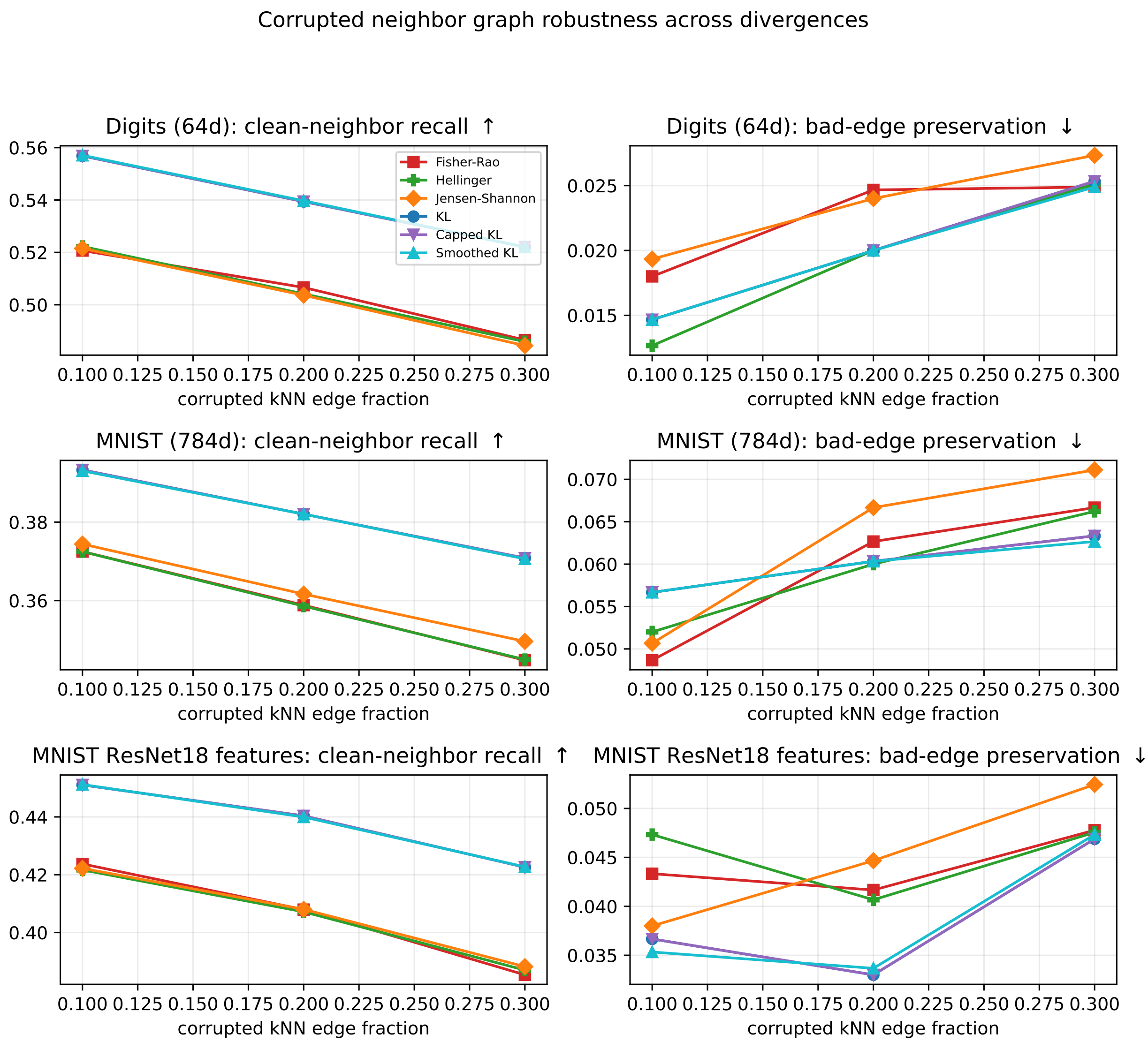

Figure 8: kNN-graph corruption benchmark on digits, MNIST, and ImageNet-pretrained ResNet18 MNIST features. The training target is a normalized corrupted neighbor graph; evaluation is against the clean feature geometry and labels.

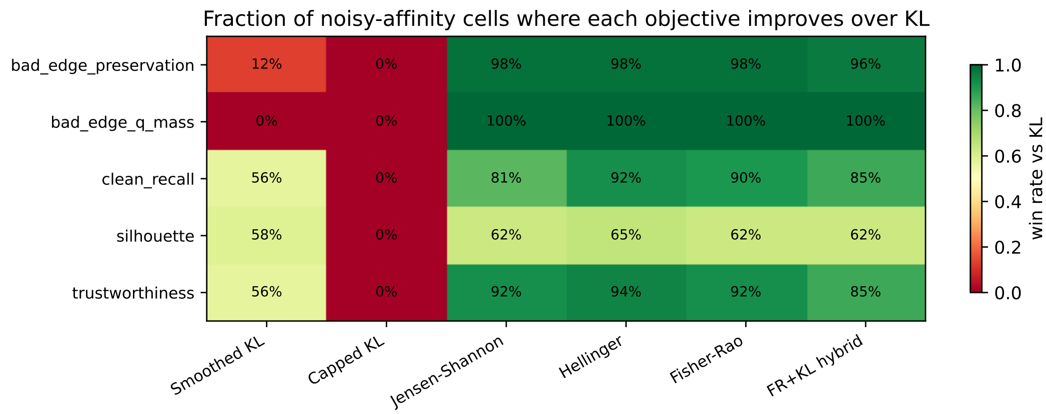

Figure 9: Win-rate heatmap for all objectives vs KL in the noisy-affinity stress test (48 nonzero cells). Jensen–Shannon and Hellinger match Fisher–Rao on the two primary robustness metrics; capped KL fails in every cell.

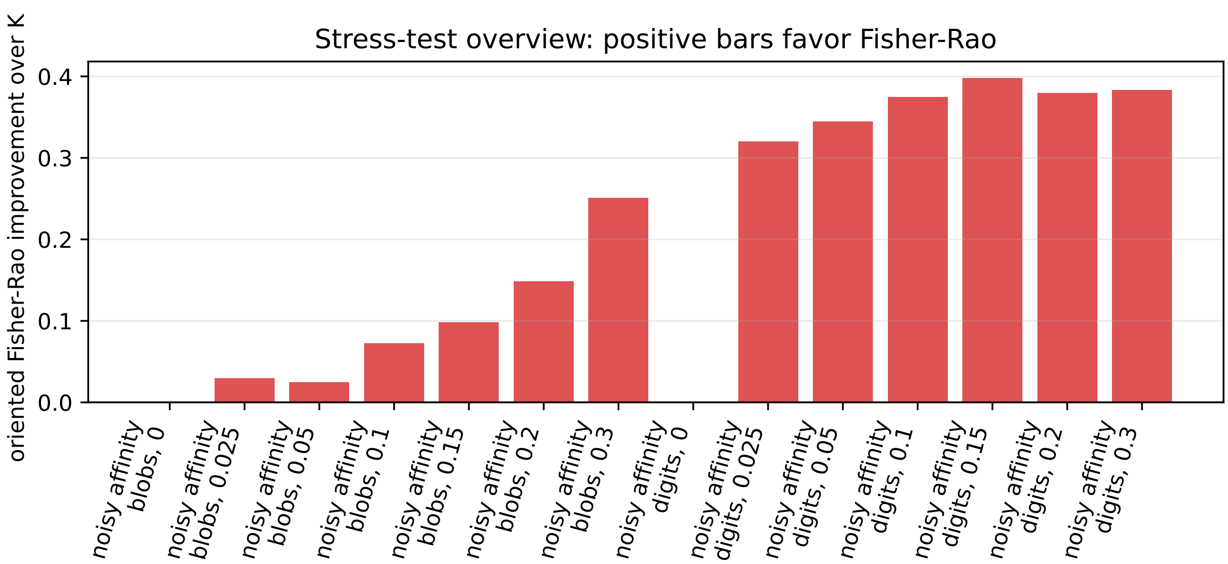

Figure 10: Stress-test overview. Each bar is oriented so positive means Fisher–Rao improves the most diagnostic metric for that stress family. The clearest signal is the noisy-affinity false-edge benchmark; other stress families are weaker or mixed.

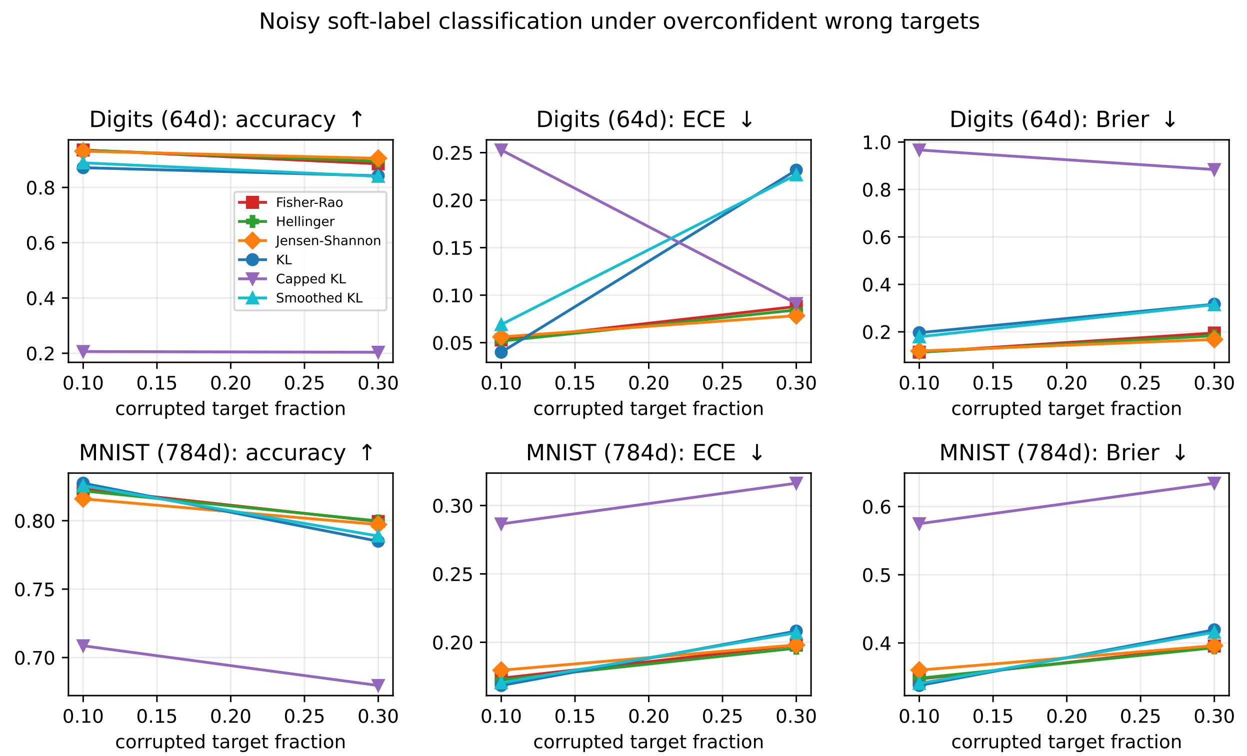

Figure 11: Noisy soft-label classification under adversarial overconfident target corruption. Fisher–Rao is most useful when KL overfits wrong high-confidence probability labels; bounded alternatives such as Hellinger and Jensen–Shannon provide important baselines.

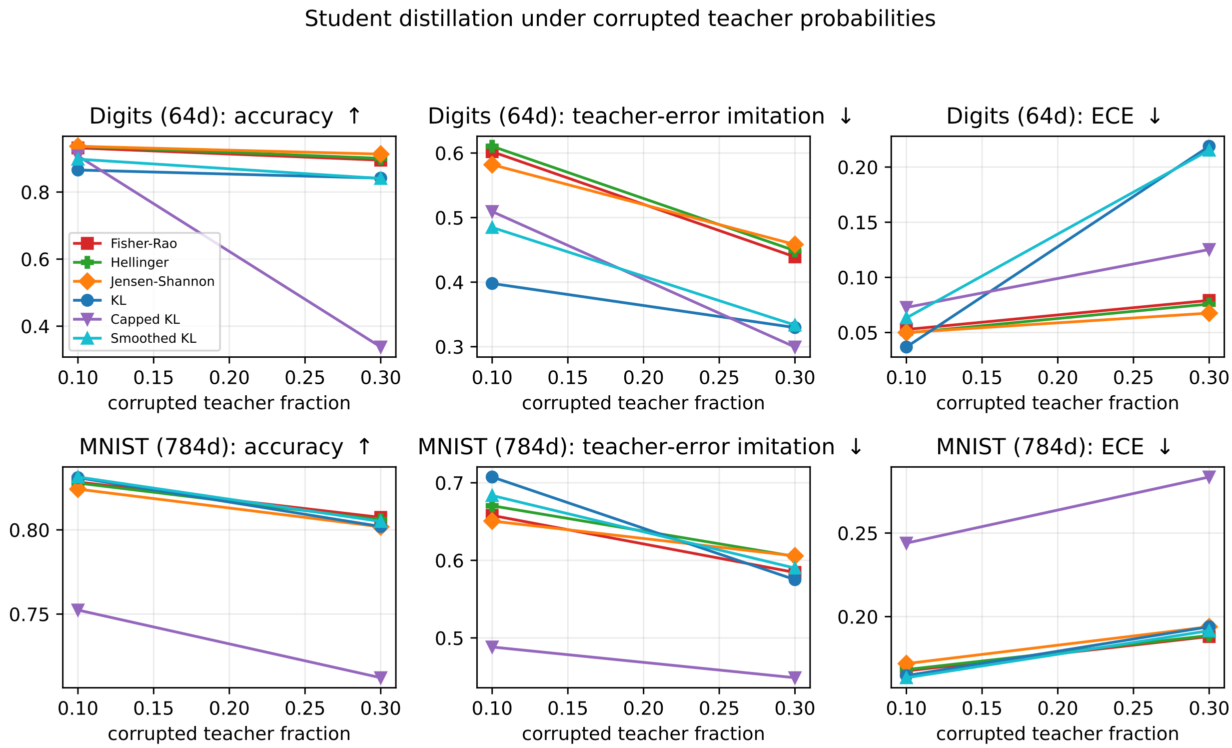

Figure 12: Corrupted teacher-student distillation. Fisher–Rao can reduce imitation of teacher mistakes, but the result is less uniform than the soft-label benchmark.

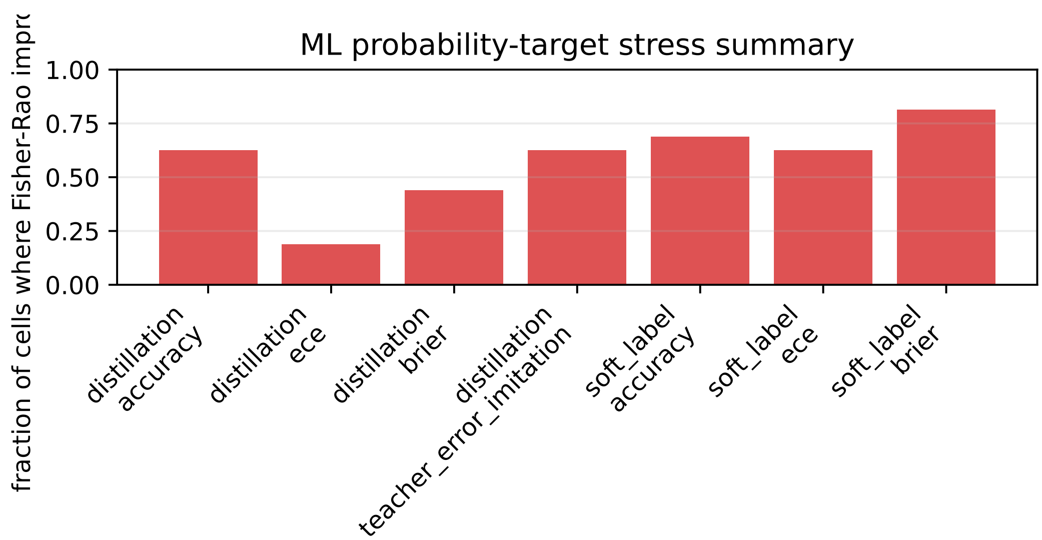

Figure 13: Fraction of nonzero corruption cells where Fisher–Rao improves over KL for the supervised probability-target stress tests.

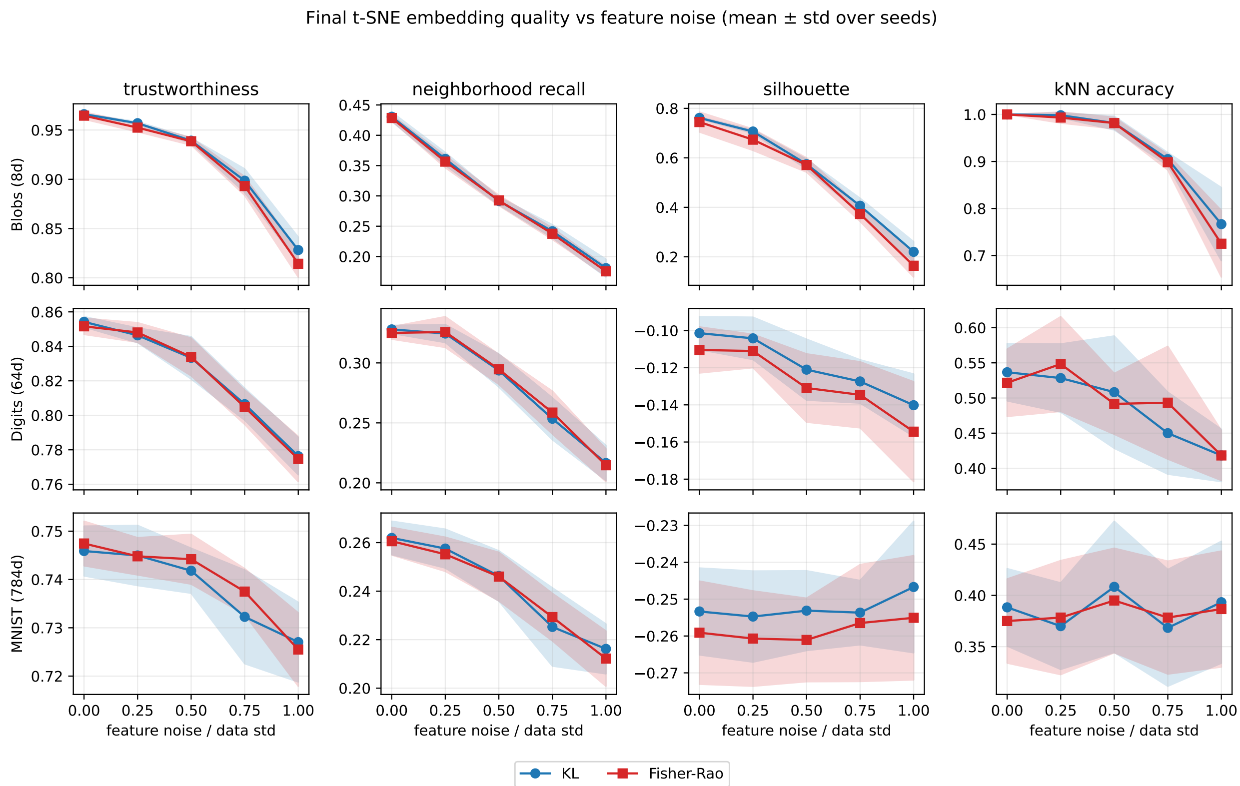

Figure 14: Final t-SNE embedding quality versus feature noise across three datasets, four evaluation metrics, two objectives, and ten seeds. Lines are seed means and shaded bands are ± 1 standard deviation. KL and Fisher–Rao track each other closely on neighborhood preservation but KL has a small consistent advantage on silhouette.

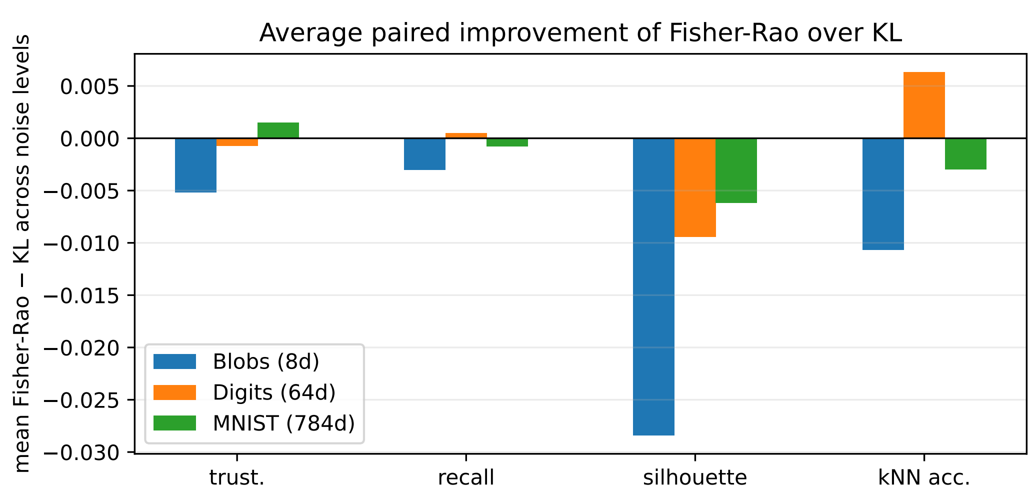

Figure 15: Average paired improvement of Fisher–Rao over KL for each metric, averaged across noise levels, broken down by dataset. Bars below zero mean KL is better on average.

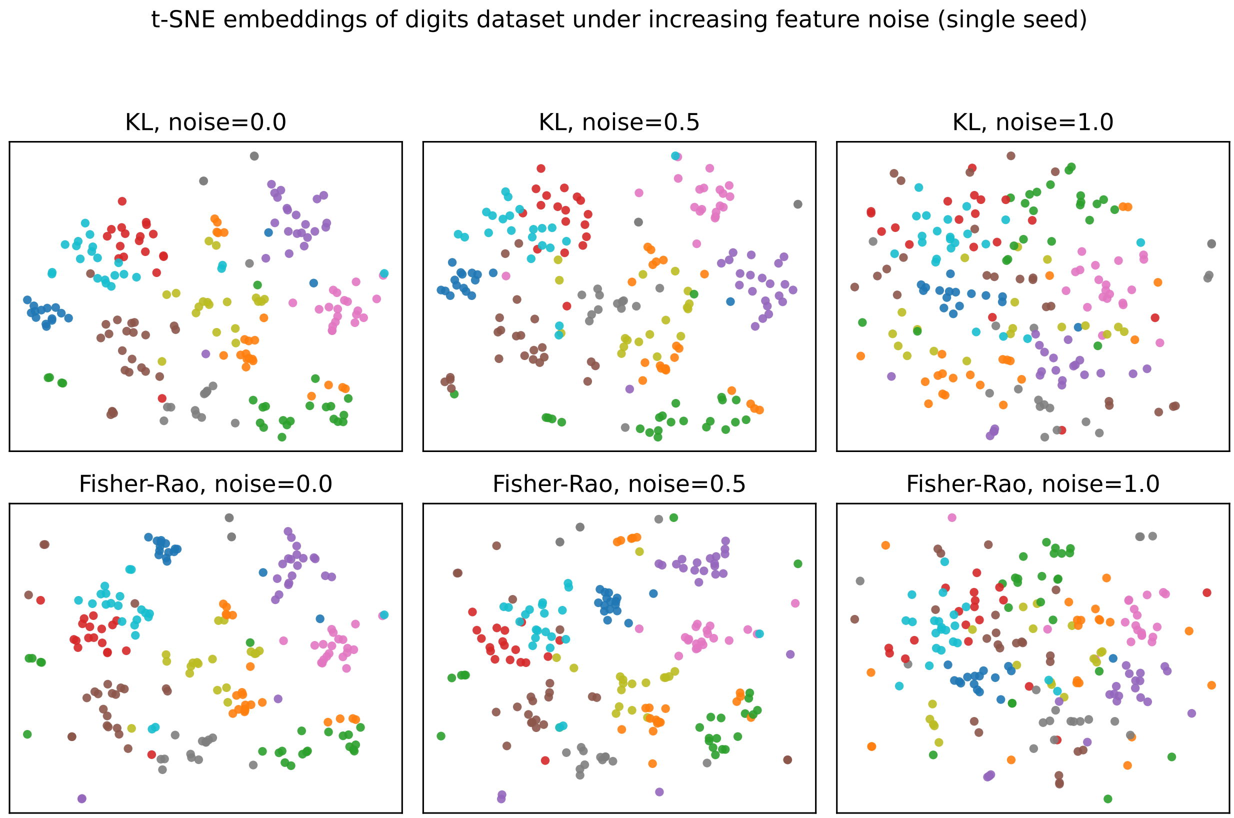

Figure 16: Qualitative t-SNE embeddings of `digits` (single seed) under three feature noise levels for the two objectives. The same initialization and optimizer are used; the two objectives produce visually similar embeddings.

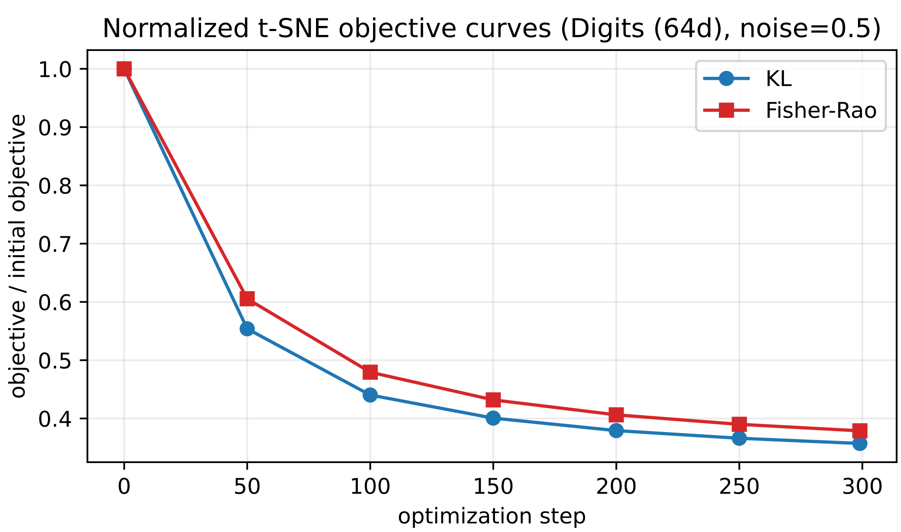

Figure 17: Normalized t-SNE objective curves on `digits` at noise $\alpha = 0.5$. Both objectives are well-behaved under Adam; the relative decrease rates differ because the underlying loss landscapes have different scales.